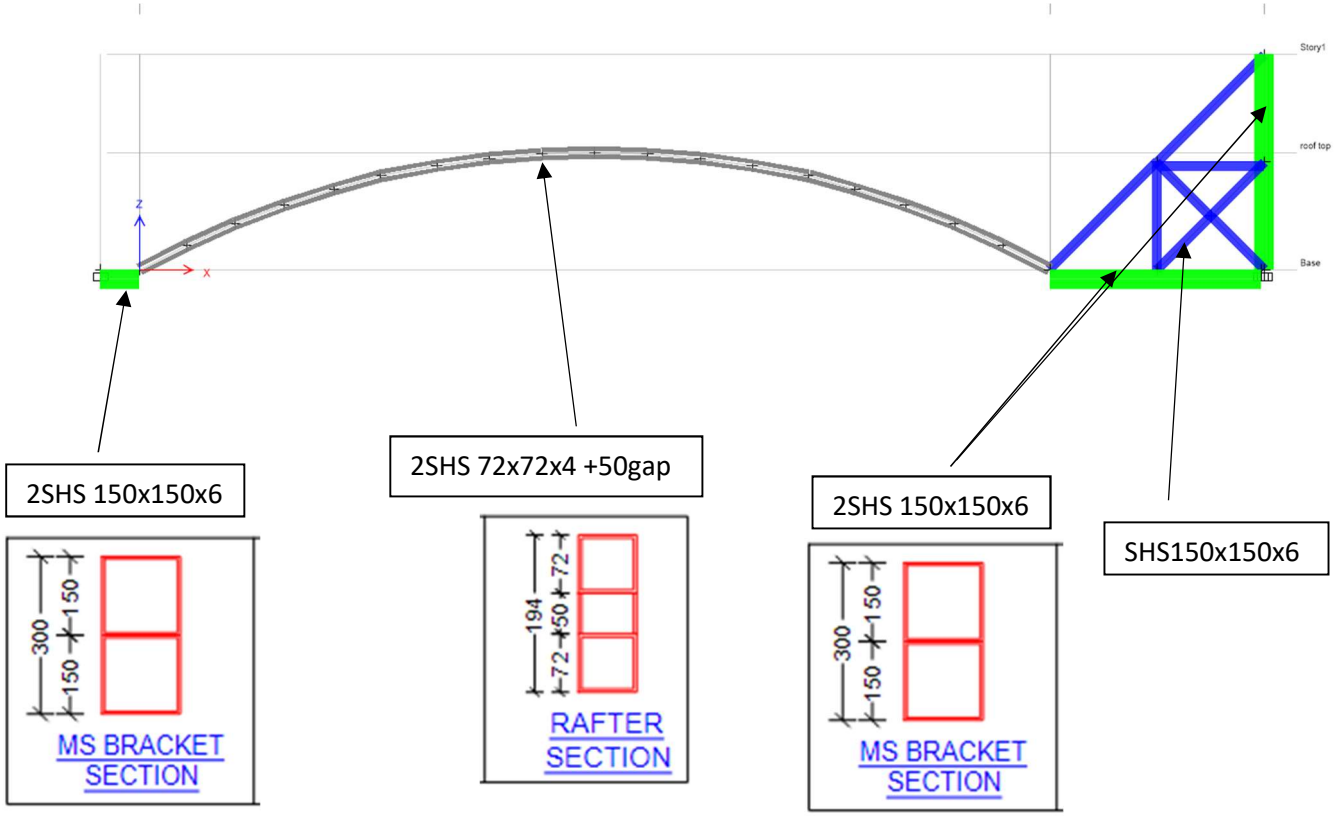
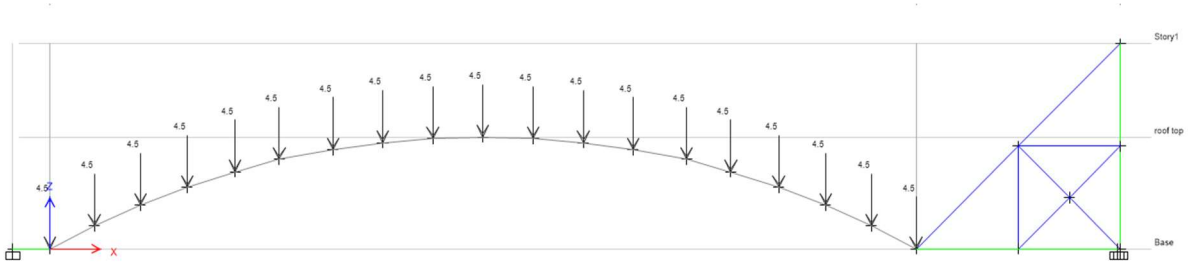
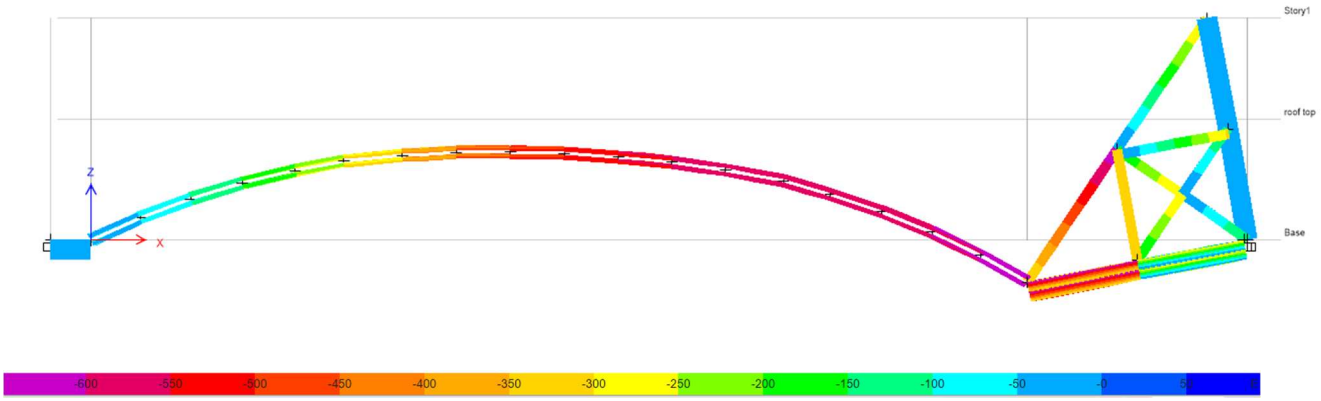


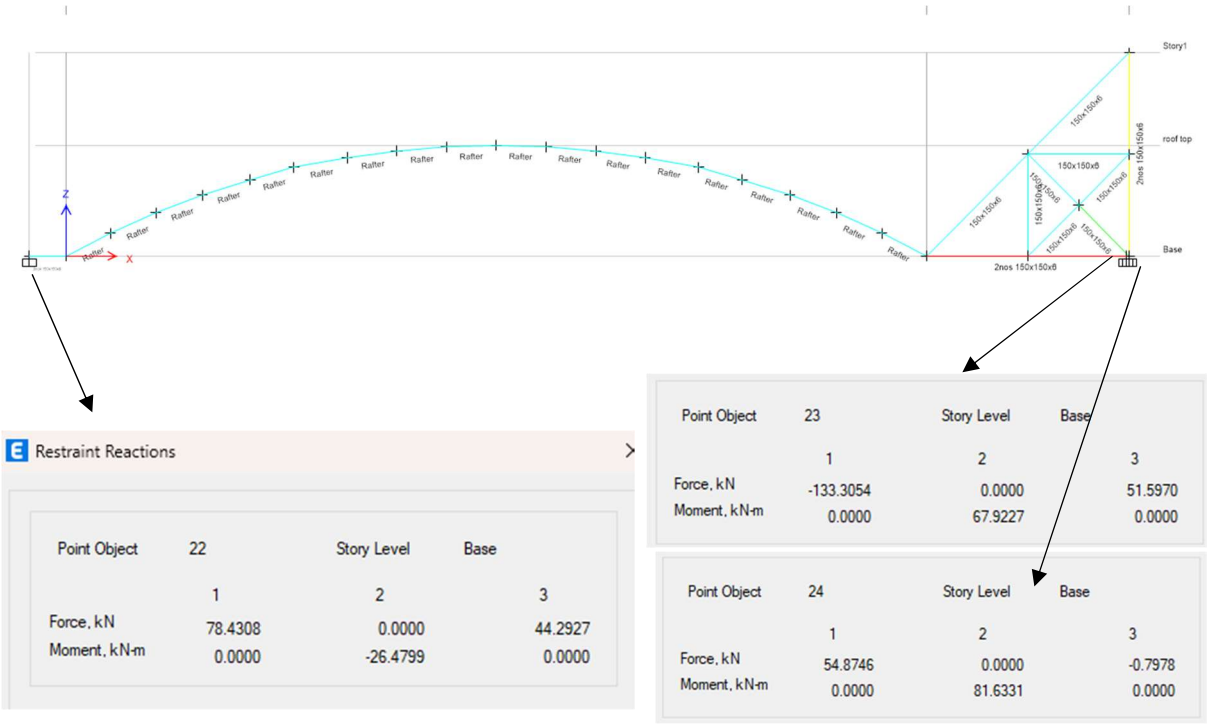
All member shall be TATA YST-310. 14m curve span with 1.8m central ridge as per vendor drawing.



Load applied as per FF= 0.25kN/sq-m and wind load is 1.5kN/sq-m



Deflection in 10^-2 mm in contour and max is 6.7mm max as coloured dark pink.

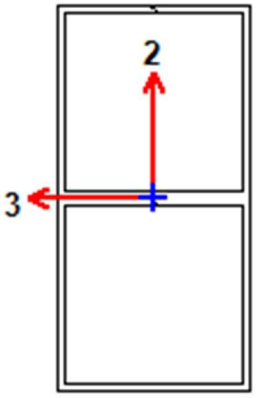


1 is horizontal in plan reaction , 2 is horizontal out of plan reaction, 3 is vertical reaction in KN

All Member Design are pass as per IS800-2007 and govern member details shown below

ETABS Steel Frame Design

IS 800:2007 Steel Section Check (Strength Summary)



Element Details (Part 1 of 2)

Level	Element	Unique Name	Location (mm)	Combo	Design Type	Element Type	Section
Base	B17	33	3250	Comb2	Beam	Special Moment Frame	2nos 150x150x6

Element Details (Part 2 of 2)

Classification	Rolled
Class 3	No

Design Code Parameters

γ_{M0}	γ_{M1}	A_n / A_g	LLRF	PLLF	Stress ratio Limit
1.1	1.25	1	1	0.75	0.95

Section Properties

A (cm^2)	I_{zz} (cm^4)	r_{zz} (mm)	$Z_{e,zz}$ (cm^3)	$A_{v,z}$ (cm^2)	$Z_{p,zz}$ (cm^3)	I_{yz} (cm^4)	I_t (cm^4)
69.1	6280.9	95.3	418.7	35.2	518.4	0	5025.9

J (cm^4)	I_{yy} (cm^4)	r_{yy} (mm)	$Z_{e,yy}$ (cm^3)	$A_{v,y}$ (cm^2)	$Z_{p,yy}$ (cm^3)	I_w (cm^6)	h (mm)
5025.9	2392.9	58.8	319.1	36.5	373.5	0	300

Material Properties

J (cm^4)	I_{yy} (cm^4)	r_{yy} (mm)
5025.9	2392.9	58.8
E (MPa)	f_y (MPa)	f_u (MPa)
210000	310	425

Stress Check Forces and Moments

Location (mm)	N (kN)	M_{zz} (kN-m)	M_{yy} (kN-m)	V_y (kN)	V_z (kN)	T_o (kN-m)
3250	-199.9581	-101.884	0	77.3954	0	0

PMM Demand/Capacity (D/C) Ratio 9.3.2.2(a)

D/C Ratio =	$P / P_{dy} + K_y * C_{my} * (M_{y,span} / M_{dy;}) + K_{LT} * (M_{z,span} / M_{dz;})$
0.994 =	0.139 + 0.856 + 0

Basic Factors

Buckling Mode	K Factor	L Factor	L Length (mm)	KL/r
Major (z-z)	1	1	3250	34.094
Major Braced	1	1	3250	34.094
Minor (y-y)	1	1	3250	55.236
Minor Braced	1	1	3250	55.236
LTB	1	1	3250	55.236

Axial Force Design

	N Force kN	T_d Capacity kN	N_d Capacity kN	P_{dy} Capacity kN	P_z Capacity kN	P_d Capacity kN
Axial	-199.9581	1947.9273	1947.9273	1441.0363	1730.3718	1441.0363

T_{dg} kN	T_{dn} kN	$N_{cr,T}$ kN	$N_{cr,TF}$ kN	A_n / A_g Unitless	N / N_d Unitless
1947.9273	2115.072	323480.8977	4695.5137	1	0.103

Design Parameters for Axial Design

	Curve	α	f_{cc} (MPa)	λ	ϕ	χ	f_{cd} (MPa)
Major (z-z)	c	0.49	1783.09	0.417	0.64	0.888	250.34
MajorB (z-z)	c	0.49	1783.09	0.417	0.64	0.888	250.34
Minor (y-y)	c	0.49	679.33	0.676	0.845	0.74	208.48
MinorB (y-y)	c	0.49	679.33	0.676	0.845	0.74	208.48

	Curve	α	f_{cc} (MPa)	λ	ϕ	χ	f_{cd} (MPa)
Torsional TF	c	0.49	679.33	0.676	0.845	0.74	208.48

Moment Designs

	M Moment kN-m	M_{span} Moment kN-m	$M_{d(yield)}$ Capacity kN-m	M_{dv} Capacity kN-m	M_{nd} Capacity kN-m	$M_{d(LTB)}$ Capacity kN-m
Major (z-z)	-101.884	-101.884	118.0054	118.0054	118.0054	118.0054
Minor (y-y)	0	0	89.9163	89.9163	89.9163	

Factors

	Curve	α_{LT}	λ_{LT}	ϕ_{LT}	χ_{LT}	C_1	M_{cr} (kN-m)
LTB	c	0.49	0.099	0.48	1	3.053	13328.4364

	C_{my}	C_{mz}	C_{mLT}	k_z	k_y	K_{LT}	M_y / M_{dy}	M_z / M_{dz}	α_1	α_2
Factors	1	0.4	0.4	1.025	1.066	0.991	0	-0.863	2	2

Shear Design

	V Force (kN)	V_d Capacity (kN)	T_o Capacity (kN-m)	Stress Ratio	Status Check
Major (y)	77.3954	594.623	0	0.13	OK
Minor (z)	0	572.4015	0	0	OK

Shear Design

	V_p (kN)	k_v (Unitless)	Λ_w (Unitless)	T_b (MPa)
Reduction	594.623	0	0	1